

1 Introduction

1.1 Definetly include

- Define masting
- Introduce different hypotheses related to masting
- Introduce that different hypotheses act on different stages of the reproductive process (e.g., flowering, pollination, seed maturation, dispersal)
- Existing studies lack emphasis on the role of reproductive traits as mechanistic links between masting and hypotheses
- Our study propose a framework connecting:
 - reproductive traits
 - stages of reproduction
 - specific masting hypotheses
- This framework can clarify which traits are under selection in masting species

1.2 Maybe include

- Justify the selection of reproductive traits (or should be in method?)
- Previous studies on relationship between seed mass and masting focus on an ecological constraint perspective, where seed mass reflects the cost of reproduction. However, we try to capture the evolutionary mechanisms underlying masting benefits. For masting to be adaptive under the predator satiation hypothesis, seeds must be consumed by animals. Therefore, fruit size might be more relevant than seed mass alone.

2 Discussion

2.1 Definetly include

- Seed mass and wind pollination are reproductive trait significantly associated with masting patterns. This might suggest that reproductive allocation plays a key role in shaping masting behavior.
- However, the fact that we didn't find relationships with other traits indicates that trait-based explanations alone are not sufficient to fully explain masting.
- **Gymnosperm are more likely to be masting species**, (is there a possibility that masting might be an ancestral strategy that becomes lost in some angiosperm lineages later?)

- **Masting is strongly related to environment as well**, some species might exhibit masting or not depending on the environment they are in, broad trait-based analysis do not capture environmental drivers, and that's the reason why we can't use any masting metrics capturing other characteristics of masting.
- We call for a better definition of masting (flower masting, seed masting)
- Different hypotheses may act at different stages, playing the selection on different species.
- Existence of clear counterexamples. Focusing on within-clade variation (oaks) may be more informative.

2.2 Maybe include

- We don't have very good data on flowering traits
- Maybe some more about data scarcity?

3 Results

3.1 Gathered data

- Number of data available for angiosperm and gymnosperm originally from book, with masting data, and the structure of data
- Angiosperm (124, 98, Y/N: (51/47); Gymnosperm (64, 61, Y/N: (57/7))
- Strong masting more common for gymnosperm
- Number of traits we have for different hypotheses for angiosperm and gymnosperm
- For angiosperm, we have data for five traits (seed weight, fruit size, oil content, seed dispersal mode, seed dormancy) related to predation satiation hypothesis; two traits (pollination mode, reproductive type) related to pollination coupling; two traits (drought tolerance, leaf longevity) related to resource matching hypothesis. For gymnosperm, we only have oil content for masting species and all the gymnosperm species are wind pollinated.

3.2 Results for each hypothesis

- Predation satiation: we found only seed weight is related to masting with having bigger seeds will increase the chance of being a strong masting species, however, the log-odds is only 0.2. We also only found this in angiosperm group.

- Pollination satiation: we found wind pollination is related to masting for angiosperm species, the log-odds for wind compare to animal or both are 1.72 and 1.23 respectively. Gymnosperm are all wind pollinated, and gymnosperm has a greater portion of masting species.
- Resource matching: We don't find any traits related to resource matching hypothesis have relationship with the masting pattern.

3.3 Multidimension-NMDS

- We don't see masting aligned well along any axis in the NMDS analysis.

3.4 Phylogenetic patterns

- First show the phylogenetic signal for masting. Masting has a strong phylogenetic signal for gymnosperm, and no phylogenetic signal for angiosperm
- Show which traits have the strongest phylogenetic pattern, (it may not be fully comparable though, not only because the way lambda is calculated for continuous traits and categorical traits are different, but also because lambda might depend on the data structure when sampling is uneven, take leaf longevity as an example).
- seed weight and pollination mode both have strong phylogenetic signal, (it is hard to tease out these traits from masting because of the phylogenetic signal, limit the inference)
- Data is strongly nested in phylogeny, and data is sparse. We have to assume the sampling is even.